2023-2024 第一学期
Advanced Mathematics 期中

重制：*cppHusky*

一、 **Question 1** [**40 marks**]

Fill in all the following blanks. Only the final results are required to be written down.

1. $\lim\limits_{x\to 0}\dfrac{\mathrm{e}^x+\ln(1-x)-1}{x-\arctan x}=$____.

2. $\lim\limits_{x\to 0}\left(\dfrac{\sin x}{x}\right)^{\frac{1}{1-\cos x}}=$____.

3. Suppose $f(x)=a^x$ $(a>0,\ a\neq 1)$, then $\lim\limits_{n\to\infty}\dfrac{1}{n^2}\ln[f(1)f(2)\ldots f(n)]=$____.

4. Suppose that $f(x)$ is derivable at $x=1$, and $\lim\limits_{x\to 1}\dfrac{f(x)+\mathrm{e}^{x-1}-3}{x-1}=-3$, then $f'(1)=$____.

5. $x=0$ is a/an ____ discontinuous point of the function $f(x)=\dfrac{\mathrm{e}^{\frac{1}{x}}-1}{\mathrm{e}^{\frac{1}{x}}+1}$. ("removable" or "jump" or "infinite" o "oscillating")

6. The equation of the tanget line to the curve $y=x^2-x-1$ perpendicular to $x+3y-1=0$ is ____.

7. Suppose $f(x)=x^2\,\mathrm{e}^{2x}$, then $f^{(10)}(0)=$____.

8. From Lagrange's theorem, we have $\sqrt{1+x}-1=\dfrac{x}{2\sqrt{1+\theta x}}$, then $\lim\limits_{x\to 0}\theta=$____.

二、 Question 2 [15 marks]

Suppose that the function $y = y(x)$ is defined by the equation $\begin{cases} x = \sqrt{1+t^2}, \\ y = \ln\left(t + \sqrt{1+t^2}\right), \end{cases}$ Find $\left.\dfrac{\mathrm{d}y}{\mathrm{d}x}\right|_{t=1}$ and $\left.\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{t=1}$.

三、 Question 3 [15 marks]

Suppose that $f(x) = \begin{cases} ax^2 + bx + c, & x \geq 0, \\ \ln(1+x), & x < 0 \end{cases}$ is derivable of order two at $x = 0$. Find a, b, c.

四、 Question 4 [15 marks]

Suppose that $y = \begin{cases} \dfrac{\ln(1-x) + a\sin x + bx^2}{x^2}, & x \neq 0, \\ \dfrac{5}{2}, & x = 0 \end{cases}$ is continuous at $x = 0$. Find a, b.

五、 Question 5 [15 marks]

Suppose $f(x)$ is continuous on $[a,\, b]$ and differentiable in $(a,\, b)$, $0 < a < b$. Prove that there exist ξ, $\eta \in (a,\, b)$, such that $f'(\xi) = \dfrac{\eta^2 f'(\eta)}{ab}$.